

Quote Placement Dominates Microstructure Prediction in Retail-Latency Market Making

A Negative Result from 70 Days of Cryptocurrency Limit Order Book Data

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Abstract

High-frequency market makers profit by acting on price signals faster than competitors. Retail and non-institutional algorithmic traders, operating under execution delays of 200 ms to 2 s, are effectively excluded from market making: by the time a quote reaches the exchange, the market has moved. A natural conjecture, and the premise of an earlier version of this work, is that machine learning models trained on limit order book (LOB) microstructure can substitute for speed by anticipating market state at order arrival.

We test that conjecture and reject it at retail-accessible data frequency. Using 269,330 LOB snapshots collected continuously over 70 days from Kraken across BTC/USD, ETH/USD, and SOL/USD, we train a survival-analysis fill probability model on 109 microstructure features and compare it against a deliberately trivial benchmark: a 32-cell lookup table containing only the mean historical fill rate for each combination of quote offset, latency window, and side. On a matched test split, the neural model attains AUC 0.6375 and the lookup table 0.6372, a difference of +0.0003. Per-asset replication reproduces the null on all three instruments. On the smallest asset, ETH/USD, the neural model *underperforms* the table by 0.0988.

We report two further findings. First, a controlled ablation shows the entire measurable predictive content of the model is carried by three scalar quantities describing the quote itself; zeroing the 109-dimensional LOB window and retaining only those scalars costs 0.085 AUC, while the reverse configuration collapses to chance (0.5034 ± 0.031 across five seeds). Second, we document a conditioning defect that invalidated our own earlier results, in which quote offset was omitted from the model input, rendering four training rows with fill rates spanning 0.782 to 0.358 numerically identical at the input layer. We report this explicitly, and treat the corrected pipeline as a reproduction control throughout.

The practical implication is direct. At roughly 90 LOB updates per hour, the rate available from a standard public exchange WebSocket feed, fill probability is a function of how far from mid a trader quotes, and microstructure signals add nothing measurable on top. This bounds where the necessary signal can be found rather than refuting latency compensation as a principle, and it indicates that the binding constraint for retail participants is data frequency rather than model capacity or sample size.

Keywords: market making, market microstructure, limit order book, fill probability, execution latency, retail trading, negative results, survival analysis

JEL: G12, G14, C45, C61, G23

1 Introduction

In competitive markets, speed is capital. Firms dominating modern equity and cryptocurrency liquidity provision have invested heavily in compressing execution latency to microseconds. Microseconds do not matter to the real economy; being first to respond to a price signal is profitable, repeatedly, at scale. This raises a precise question: is speed itself the valuable resource, or is it a proxy for the ability to act on information before it becomes stale?

If the latter, speed might not be strictly necessary. A participant who cannot act in microseconds could still act correctly, provided they can predict what the market will look like when their order arrives. Version 1 of this work proposed exactly that: *latency-compensating market making*, in which deep learning models trained on LOB microstructure substitute predictive accuracy for execution speed.

This version reports that the conjecture does not survive contact with the data we are able to collect, and characterises precisely why.

1.1 What changed

Version 1 presented a theoretical framework alongside results on synthetically generated LOB data. Those synthetic results were encouraging. They were also, we now recognise, uninformative: a geometric Brownian motion price process with a hand-specified fill rule contains whatever structure the specification puts there, and the fill probability model was in effect recovering its own generator.

On real data the picture inverted. Our first real-data results were worse than chance, which prompted a diagnostic that located a defect in the model’s input pipeline (Section 6). Correcting it produced a genuine improvement. But a like-for-like comparison against a trivial baseline (the comparison we should have run first) showed that the corrected model does not outperform a lookup table on three variables.

We report the sequence in full, including our own error, because the error is instructive: a model can appear to improve with more data, respond to ablation as though its features matter, and still carry no information beyond a trivial benchmark that was never run.

1.2 Contributions

1. We formalise the latency-compensating market making problem as an extension of Avellaneda and Stoikov [2008] incorporating execution delay (Section 3). This framework is unchanged from Version 1 and we believe it remains correct; our negative result concerns its empirical instantiation, not its logic.
2. We release 269,330 timestamped 10-level LOB snapshots across three cryptocurrency pairs, collected continuously over 70 days, together with the complete collection and feature pipeline (Section 4).
3. We show that a survival-analysis fill probability model over 109 microstructure features does not outperform a 32-cell lookup table on quote offset, latency, and side, either pooled (+0.0003 AUC) or per-asset (Section 7).
4. We isolate the source of predictive content via a three-arm ablation, showing that quote geometry accounts for essentially all of it (Section 7.4).
5. We document a model conditioning defect that invalidated our own prior results, and use its deliberate reproduction as an experimental control (Section 6).

1.3 What this result does and does not establish

Because negative results are easy to over-read, we state the scope precisely.

Established. At an observed update frequency of approximately 90 LOB snapshots per hour per asset (the rate delivered by Kraken’s public WebSocket book channel at depth 10), a 90,754-parameter LSTM over 109 engineered microstructure features provides no measurable improvement in fill prediction over quote offset, latency window, and side alone, on any of three cryptocurrency pairs.

Not established. We have *not* shown that LOB microstructure is uninformative about fill probability in general. A substantial literature demonstrates otherwise at higher frequencies [Maglaras et al., 2022, Arroyo et al., 2024]. Our data are sampled at a median inter-update interval of 18.2 seconds (mean 40.1 seconds), which is three to four orders of magnitude coarser than the tick-level data used in that work. The most probable explanation for our null is that microstructure signal exists but does not survive this sampling interval. We regard distinguishing that explanation from the alternatives as the central open question (Section 8).

Not tested. The reinforcement learning agent proposed in Version 1 consumed fill probability estimates from the defective model and was trained only on synthetic data. We therefore report no RL results in this version. See Section 7.5.

2 Related Work

2.1 Market making theory and latency

The modern framework originates with the inventory models of Garman [1976] and Ho and Stoll [1981], formalised as a stochastic control problem by Avellaneda and Stoikov [2008], who derived closed-form optimal quotes as functions of inventory, risk aversion, and order arrival intensity. Extensions incorporate adverse selection and execution uncertainty [Cartea et al., 2018], and Cartea and Sánchez-Betancourt [2023] introduce stochastic delay into optimal execution.

Nearly all analytical market making models assume negligible execution latency. Lehalle and Mounjid [2022] show this assumption is load-bearing rather than cosmetic: the value of predictive signals about future order flow is systematically eroded by latency, since a trader who can predict adverse flow but cannot act quickly enough derives diminishing benefit. Our result is consistent with theirs and sharpens it. In our setting the erosion is complete at the data frequency available to us.

2.2 The latency arms race

Moallemi and Saglam [2013] provide the first formal model of latency cost. Budish et al. [2015] characterise continuous-time limit order books as generating structurally wasteful speed investment. Aquilina et al. [2022] quantify this empirically, finding latency arbitrage races occur roughly once per minute per symbol in FTSE 100 stocks with modal durations of 5–10 microseconds, and estimating that eliminating them would reduce effective spreads by approximately 17%.

Proposed remedies (the IEX speed bump of Hu, 2019, frequent batch auctions in Budish et al., 2015, micro-burst fees in Brolley and Zoican, 2023) operate at the exchange level and require regulatory or venue adoption. None offers recourse to an individual trader on existing infrastructure, which was the motivation for pursuing a participant-level technical approach.

2.3 Fill probability estimation

Lo et al. [2002] established econometric foundations for execution probability modelling. Maglaras

et al. [2022] introduced a deep learning approach treating execution as a time-to-fill distribution. Arroyo et al. [2024] advanced this with a convolutional-transformer architecture using survival analysis, which handles the right-censoring inherent in limit order data where many orders are cancelled before executing.

Our fill probability model follows Arroyo et al. [2024] in its survival formulation. The critical difference in our setting is data frequency: that work uses tick-level message data, whereas we use snapshot data at a median 18.2 second interval. We return to this in Section 8, as we believe it is the primary explanation for our divergent result.

2.4 Reinforcement learning for market making

Spooner et al. [2018] demonstrated RL agents learning competitive market making policies in simulation; Guo et al. [2023] added continuous action spaces and attention-based LOB features; Briola et al. [2021] established viability in high-frequency cryptocurrency settings. The most directly relevant work is RELAVER [Jiang et al., 2025], which explicitly models random execution delays of 30–100 ms with a batch matching mechanism.

Version 1 of this paper positioned itself as extending RELAVER to a retail latency regime an order of magnitude larger. We no longer make that claim, as the predictive component such an extension would require does not function at our data frequency.

3 Problem Formalisation

We retain the formalisation from Version 1, as the negative result concerns its empirical instantiation rather than its structure.

3.1 Standard setup

A market maker operates over horizon $[0, T]$. The mid-price follows $S_t = S_0 + \sigma W_t$ with $\sigma > 0$ and W_t standard Brownian motion. Posting a bid at distance δ^b and ask at δ^a from mid, fills arrive as Poisson processes with intensities

$$\lambda^b(\delta^b) = Ae^{-k\delta^b}, \quad \lambda^a(\delta^a) = Ae^{-k\delta^a}, \quad (1)$$

where $A, k > 0$ govern the decay of arrival rate with distance from mid. Inventory q_t evolves with fills and wealth X_t accumulates spread capture net of inventory risk. The objective is

$$\max_{\delta^b, \delta^a} \mathbb{E}[X_T + q_T S_T - \gamma q_T^2], \quad (2)$$

with $\gamma > 0$ the inventory risk aversion parameter.

3.2 Introducing execution latency

We depart from the standard framework by introducing execution latency ℓ , the delay between quoting decision and order arrival at the exchange. We consider $\ell \in [200 \text{ ms}, 2000 \text{ ms}]$, characterising traders without co-location infrastructure.

Under latency ℓ , a quote posted at time t arrives at $t + \ell$ into an order book state $\mathcal{F}_{t+\ell}$ that is unobserved at decision time. The relevant fill probability is

$$P^{\text{fill}}(\delta, \ell, \mathcal{F}_t) = \mathbb{P}[\text{order fills} \mid \delta, \mathcal{F}_{t+\ell}], \quad (3)$$

which depends on a future state. This creates two compounding losses relative to the zero-latency benchmark: *adverse selection amplification*, where quotes are picked off when the mid moves against them during transmission, and *cancellation impotence*, where a trader detecting an adverse signal cannot cancel before being filled.

3.3 The latency-compensating problem

Let $\hat{P}^{\text{fill}}$ denote an estimator of fill probability and $\hat{S}_{t+\ell}$ an estimator of the mid-price at arrival. The latency-compensating problem is

$$\max_{\delta^b, \delta^a} \mathbb{E} \left[X_T + q_T \hat{S}_T - \gamma q_T^2 \mid \hat{\mathcal{F}}_{t+\ell} \right] \quad (4)$$

subject to the participation constraints

$$\hat{P}^{\text{fill}}(\delta, \ell, \mathcal{F}_t) \geq \tau_{\text{fill}}, \quad \left| \hat{S}_{t+\ell} - S_t \right| \leq \Delta_{\text{safe}}, \quad (5)$$

with abstention whenever either constraint is violated.

3.4 Viability conditions

Version 1 stated three conditions for viability. We restate them, since our result speaks directly to the first.

Condition 1 — Predictive accuracy. The estimators $\hat{P}^{\text{fill}}$ and $\hat{S}_{t+\ell}$ must be accurate enough that the constraints in (5) select genuinely favourable states rather than arbitrary ones.

Condition 2 — Uninformed order flow. The market must contain sufficient noise-trader flow to generate fill opportunities that are not systematically adverse [Chen and Moallemi, 2018].

Condition 3 — Abstention discipline. The abstention rule must be strictly enforced; a strategy that quotes indiscriminately degrades to the latency-penalised baseline.

Our empirical finding is that Condition 1 fails at retail-accessible data frequency. The failure is not that no estimator achieves discrimination. It is that the achievable discrimination is entirely reproduced by a lookup table on quote geometry, which requires no model and confers no informational advantage. Conditions 2 and 3 are untested here.

4 Data

4.1 Collection

LOB data were collected continuously from Kraken via WebSocket API v2, subscribing to the `book` channel at depth 10 for BTC/USD, ETH/USD, and SOL/USD. Each update is timestamped at receipt and stored with the full 10-level bid and ask ladder in Parquet format, partitioned by asset, date, and hour.

Collection ran on a cloud virtual machine from 15 June to 24 August 2026. The collector is managed by a systemd unit with `Restart=always`; over the final three-week window it was restarted automatically three times following WebSocket disconnections. An earlier configuration using a detached terminal session and a bounded reconnect loop failed silently on 20 July and remained down for fourteen days, costing an estimated 90,000 snapshots. We note this because unattended collection reliability is a material and under-discussed constraint on independent microstructure research.

4.2 Sampling characteristics

Kraken’s book channel is event-driven: updates are emitted when the visible book changes, not on a fixed schedule. The resulting sampling is both sparse and irregular, and this is central to interpreting our results.

Table 1: Inter-update intervals, BTC/USD.

Statistic	Value
Mean gap	40.1 s
Median gap	18.2 s
Minimum gap	<0.01 s
Maximum gap	6339 s
Updates per hour	≈ 90

The maximum gap corresponds to a collector outage rather than market quiescence. The mean-median divergence indicates a heavy right tail: most updates cluster, with long quiet intervals between bursts.

This sampling rate is the single most important feature of our dataset. A latency window of 200 ms is roughly one ten-thousandth of the mean interval between observations. The model is therefore asked to discriminate outcomes at a horizon far shorter than its own sampling resolution. We return to this in Section 8.

Table 2: Collected dataset, 15 June – 24 August 2026.

Asset	Snapshots	Label rows	Overall fill rate
BTC/USD	111,002	3,551,392	0.734
ETH/USD	34,380	1,099,488	0.754
SOL/USD	123,948	3,965,664	0.534
Total	269,330	8,616,544	—

ETH/USD yields substantially fewer snapshots over an identical collection window. The visible book on this venue is genuinely less active; this is not a collection artefact. The asymmetry matters for Section 7.3.

4.3 Label construction

For each snapshot we simulate limit order submissions at quote offsets $\delta \in \{1, 2, 3, 5\}$ ticks on each side, and record whether each simulated order would have filled within each latency window $\ell \in \{200, 500, 1000, 2000\}$ ms based on subsequent book evolution. This yields $4 \times 4 \times 2 = 32$ label rows per snapshot. Orders not filled within ℓ are treated as right-censored at ℓ .

Table 3 reports marginal fill rates. Both variables show strong, monotone structure, which is what makes the lookup-table baseline of Section 7.1 a demanding comparison rather than a straw one.

Table 3: Marginal fill rates by quote offset and latency window, SOL/USD.

Offset (ticks)	Fill rate	Latency (ms)	Fill rate
1	0.782	200	0.385
2	0.550	500	0.498
3	0.446	1000	0.585
5	0.358	2000	0.669

4.4 Features

From each snapshot we construct 109 features: 40 raw price and volume levels, normalised prices and volumes, mid-price, spread, spread in basis points, order book imbalance, depth imbalance, and rolling statistics and lagged differences of these quantities. Prices are normalised by contemporaneous mid; volumes by a rolling mean.

Two pipeline parameters required adjustment for sparse data. The rolling normalisation window was reduced from 1000 to 10 snapshots, and lag features from $\{1, 5, 10, 20, 50\}$ to $\{1, 2, 3, 5\}$; at the original settings, nearly all rows were eliminated by the subsequent `dropna`, which we replaced with zero-fill. These changes are necessary consequences of the sampling rate: a 1000-snapshot rolling window spans roughly eleven hours of wall-clock time at 90 updates per hour, which is not a meaningful local normalisation.

5 Fill Probability Model

5.1 Survival formulation

We model time-to-fill $T \geq 0$ with survival function $S(t) = \mathbb{P}[T > t]$ and hazard $h(t)$. The fill probability within the latency window is

$$\hat{P}^{\text{fill}}(\delta, \ell) = 1 - S(\ell) = \mathbb{P}[T \leq \ell]. \quad (6)$$

Observations are right-censored: we observe $\tilde{T} = \min(T, C)$ with censoring indicator Δ . Models are trained by minimising the negative log-likelihood under censoring,

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \left[\Delta_i \log h(\tilde{T}_i) + \log S(\tilde{T}_i) \right]. \quad (7)$$

5.2 Architecture

The model encodes a lookback window of $L = 50$ snapshots with a two-layer LSTM (hidden dimension 64, dropout 0.2), producing a representation h . A conditioning vector

$$c = [\ell/\ell_{\max}, \delta/\delta_{\max}, \text{side}] \in \mathbb{R}^3 \quad (8)$$

is passed through a small MLP and concatenated with h before a monotone survival decoder with K discrete hazard outputs. Total parameters: 90,754. Training uses Adam at $\eta = 10^{-3}$ with early stopping on validation loss.

The composition of c is the subject of Section 6.

5.3 Evaluation

We report two ranking metrics that answer different questions, and we report both throughout.

Fill AUC. Area under the ROC curve for the binary filled/not-filled task, computed exactly via the Mann-Whitney rank-sum identity with ties credited 0.5. This measures whether the model ranks orders that filled above orders that did not.

Concordance index. Computed on filled orders only, measuring whether predicted fill probability ranks consistently with realised fill *time*.

AUC is the metric of primary interest, because the abstention rule in (5) tests $\hat{P}^{\text{fill}} \geq \tau_{\text{fill}}$, a binary fill/no-fill decision that never compares timings. We fixed this choice on the argument above, prior to observing that AUC would be the *lower* of the two numbers on our data; we report both regardless.

6 A Conditioning Defect and Its Use as a Control

Our first real-data results were substantially worse than chance. Our initial hypothesis was insufficient data. That hypothesis was wrong, and the diagnostic that displaced it is worth describing, because both the error and its detection are informative.

6.1 The defect

The conditioning vector c originally contained only ℓ . Quote offset δ and side were used to construct labels but never reached the model.

The consequence is structural. For a fixed snapshot and latency, the eight label rows spanning four offsets and two sides present *identical* model input while carrying different targets, with marginal fill rates ranging from 0.782 at 1 tick to 0.358 at 5 ticks (Table 3). No function of the input can separate them. The loss-minimising response is to predict the conditional mean and ignore the input entirely, which is what the model did: validation loss moved by roughly 0.003 over 40 epochs across every run.

This also explains an observation that had misled us. Across successive collection milestones, measured C-index moved 0.5008, 0.5308, 0.5526 as the dataset grew from 21K to 51K to 103K snapshots. We read this as convergence toward a target with increasing data. It was not: all three values lie within seed variance of chance, and the apparent trend was noise. A subsequent run on 269K snapshots returned 0.4749, worse with more data. That is the expected behaviour when additional data supplies additional mutually contradictory input-target pairs.

6.2 Correction and control

Adding δ and side to c raised C-index from 0.4749 to 0.6266 on an otherwise identical configuration, and validation loss movement increased roughly eightfold (0.003 to 0.023).

We retain the defective configuration as an experimental arm, denoted LOB, in which the LOB window is supplied but δ and side are reset to constants. It is a reproduction control. Any claim that a result derives from LOB features should come with a demonstration that the corresponding defective configuration performs at chance. In our ablation it does, at 0.5034 ± 0.031 across five seeds (Section 7.4).

6.3 Consequence for prior results

All real-data figures reported in Versions 1.0–1.7 of this paper were produced by the defective configuration and should be disregarded. This includes the C-index progression cited above and the projection that approximately 300,000 snapshots would be required for convergence, which was extrapolated from noise.

7 Results

7.1 The baseline

We compare the model against a lookup table: the mean fill rate for each $(\delta, \ell, \text{side})$ cell, 32 cells in total, estimated on training data and applied unchanged to test data. It has no learned parameters, no access to the order book, and no temporal information.

We emphasise that this is a demanding baseline rather than a straw one. The marginals in Table 3 show quote offset spanning a 0.42 range in fill rate and latency spanning 0.28. Any model that fails to beat this table is failing to add value over two variables the trader chooses directly and already knows.

7.2 Pooled result

Table 4: LSTM versus lookup table, pooled across three assets, matched test split ($n = 29,983$).

Predictor	Parameters	AUC
Lookup table (32 cells)	32	0.6372
LSTM (109 LOB features)	90,754	0.6375
Difference		+0.0003

The difference is three ten-thousandths of an AUC point. For reference, seed variance alone on this configuration is ± 0.034 , two orders of magnitude larger. The neural model and the lookup table are, on this evidence, indistinguishable.

7.3 Per-asset replication

We repeated the comparison independently per asset, training separate models and fitting separate tables, with three seeds each. Decision thresholds were fixed before running: a gap exceeding +0.02 would indicate LOB signal; within ± 0.02 , none.

Table 5: Per-asset comparison, AUC, three seeds per asset.

Asset	Snapshots	Lookup	LSTM (mean $\pm$ sd)	Gap
BTC/USD	111,002	0.6396	0.6379 ± 0.006	−0.0017
ETH/USD	34,380	0.6510	0.5522 ± 0.005	−0.0988
SOL/USD	123,948	0.6821	0.6739 ± 0.012	−0.0082

No asset shows LOB signal. Three observations follow.

First, SOL/USD carries the highest lookup AUC in the study, 0.6821. Its fill rate falls from 0.654 at 1 tick to 0.251 at 5 ticks at 200 ms latency, a far steeper gradient than BTC or ETH. We had hypothesised this thinner, more competitive book would be where microstructure mattered most. Instead the steepness is fully captured by quote geometry, leaving nothing for the model to add.

Second, ETH/USD is a genuine failure rather than a null: the model underperforms the table by 0.099, consistently across seeds (sd 0.005). Its training runs terminated at epochs 12, 9, and 29 with validation loss moving only 1.7437 to 1.7316. With 34,380 snapshots, roughly a third of the other assets, the model is undertrained and actively discards information the table retains. We report this rather than tuning it away, as it bounds the sample size below which this architecture is counterproductive.

Third, the pooled null is not an artefact of averaging heterogeneous assets. It holds separately on each.

7.4 Ablation

To locate the source of predictive content we ran three arms with identical architecture, data, and schedule, varying only the input:

full LOB window plus $[\ell, \delta, \text{side}]$.

cond LOB window zeroed; conditioning retained.

lob LOB window retained; δ , side reset to constants (the defective configuration of Section 6).

Table 6: Ablation, C-index, five seeds per arm.

Arm	Mean	SD	Min	Max
FULL	0.6101	0.034	0.5670	0.6536
COND	0.5247	0.041	0.4853	0.5885
LOB	0.5034	0.031	0.4678	0.5427

FULL exceeds COND by 0.085, approximately 2.3 pooled standard deviations, with ranges that barely overlap. Read alone, this appears to show LOB features contributing.

It does not, and the distinction matters. COND is a model trained with its dominant input stream zeroed, a damaged network rather than a fair alternative predictor. The lookup table comparison in Table 4 is the correct counterfactual, and it shows no gain. The ablation establishes that the model’s internal representation depends on the LOB window; it does not establish that the window carries information unavailable elsewhere. We flag this because the ablation was run first, and taken alone would have supported the opposite conclusion.

LOB at 0.5034 confirms the defect reproduces to chance across five seeds.

7.5 Status of the reinforcement learning component

Version 1 proposed a latency-aware PPO agent consuming fill probability estimates, with a reward function penalising adverse fills attributable to execution delay. We report no RL results in this version, for two reasons.

The agent was trained only on synthetic data, where it did not converge: per-episode PnL oscillated without trend across 2,000 episodes. It also consumed fill probability estimates from the defective model, which makes those runs uninterpretable regardless of the environment.

The reward function itself remains, in our view, a reasonable contribution, but it is untested, and we now present it as a proposal rather than a validated component. Given the results above, an agent conditioned on our fill probability estimates would receive no information beyond quote offset and latency, both of which it already selects directly.

8 Interpretation

Three explanations are consistent with our results. We cannot fully distinguish them, and say so.

8.1 Sampling frequency (most likely)

At a median 18.2-second inter-update interval, a 200 ms latency window is roughly 1/90th of the typical gap between consecutive observations. Nearly all short-horizon dynamics (queue position changes, fleeting imbalance, order flow bursts) occur entirely between our samples and are unobservable.

This is consistent with the wider literature. Arroyo et al. [2024] and Maglaras et al. [2022] demonstrate meaningful fill prediction from LOB data at tick resolution, three to four orders of magnitude finer. Our null does not contradict those findings. It locates a frequency below which the signal they exploit is no longer present.

If correct, this has a direct practical implication that inverts the framing of Version 1. The binding constraint for a retail participant is data frequency rather than model sophistication or dataset size. And data frequency is purchasable: full-depth or tick-level feeds are commercially available. The democratisation argument weakens accordingly, since the barrier turns out to be another form of infrastructure cost rather than something a model can substitute for.

8.2 Feature construction

Our 109 features are standard but were computed under parameters adapted for sparse data (Section 4). A rolling window of 10 snapshots spans roughly seven minutes of wall-clock time here. It is possible that better-designed features (explicitly time-aware rather than count-based, or incorporating trade prints alongside book snapshots) would recover signal this construction destroys. We consider this less likely than the frequency explanation, since it would have to overcome a gap of 0.0003, but we have not excluded it.

8.3 Architecture and objective

The survival formulation with 90,754 parameters may be poorly matched to this data. The ETH result, where the model actively underperforms a table, shows the architecture can destroy information at small sample sizes. A simpler approach, gradient boosting on flat features being the obvious candidate, might extract more. We regard this as a genuine gap in the present work: we did not run that comparison, and it is the first thing a referee should ask for.

9 Limitations

We state the limitations of the negative result directly, since a null claim is only as strong as the effort made to overturn it.

Single split. All results use one 70/15/15 temporal split. Repeated seeds capture initialisation and batch-order variance but not sampling variance. A different split might yield different numbers,

though the $+0.0003$ pooled gap would need to move by two orders of magnitude to change the conclusion.

One architecture. We tested an LSTM survival model. The convolutional-transformer of Version 1 was not re-evaluated after the conditioning fix, and no gradient-boosted or linear baseline was run. We can say this architecture does not beat a lookup table; we cannot say no model would.

One venue, one feed type. Kraken’s depth-10 book channel. Neither other exchanges nor other feed types (trade prints, full-depth, message-level) were tested, and the frequency hypothesis in Section 8 predicts precisely that a different feed would give a different answer.

Simulated fills. Labels come from replaying subsequent book states, not from executed orders. Queue position is not modelled, so we implicitly assume a quote at a price level fills when that level trades through, which is optimistic for a latecomer to the queue. Whether this biases against microstructure features specifically is unclear.

ETH undertrained. We report the ETH result as-is rather than tuning it. It should be read as a statement about this architecture at this sample size, not about ETH/USD.

No live trading. Everything here is offline evaluation. No capital was deployed and no strategy is validated for deployment.

10 Implications

10.1 For researchers

Report the trivial baseline. Our ablation showed a clean $+0.085$ separation and would, presented alone, have supported the opposite conclusion of this paper. The lookup table cost two minutes to implement and reversed the finding. In microstructure prediction, where a small number of variables often carry most of the mechanically available signal, an ablation against a damaged version of one’s own model is not sufficient evidence that features matter.

Sampling frequency deserves explicit treatment. Papers reporting LOB prediction results do not consistently state inter-update intervals, yet our results suggest this parameter can determine whether an effect exists at all.

10.2 For retail practitioners

The direct implication is favourable in one narrow sense: if fill probability at these frequencies is a function of quote offset, latency, and side, it can be estimated with a lookup table from a few weeks of data. No GPU, no deep learning, no ML expertise. The model was not necessary.

The broader implication is unfavourable. Version 1 argued that AI could substitute for infrastructure and thereby democratise liquidity provision. Our results indicate that at the data frequency available for free, there is no microstructure edge to extract. Accessing finer data means paying for it, which reintroduces the infrastructure barrier the framework was meant to circumvent.

10.3 For market design

Our finding is orthogonal to the case for batch auctions and speed bumps [Budish et al., 2015, Aquilina et al., 2022], which rest on the social cost of the speed race rather than on whether individual participants can compensate technologically. If anything it strengthens the case for venue-level reform: we find no evidence that participant-level technical adaptation is a viable substitute at retail data access, which removes one argument against structural intervention.

We are more cautious than Version 1 regarding AI-driven democratisation of market microstructure. That version asserted the structural concentration of liquidity provision is “a solvable problem, and artificial intelligence is part of the solution.” We have no evidence for that claim and some against it.

11 Conclusion

We set out to test whether machine learning models trained on limit order book microstructure could substitute for execution speed in market making, enabling participation by traders operating at 200 ms–2 s latencies. Using 269,330 LOB snapshots collected over 70 days across three cryptocurrency pairs, we find they do not, at the data frequency available from a public exchange feed.

A survival-analysis model over 109 microstructure features matches a 32-cell lookup table on quote offset, latency, and side to within 0.0003 AUC pooled, and fails to beat it on any of three assets individually. A controlled ablation places essentially all measurable predictive content in quote geometry. A deliberately reproduced conditioning defect performs at chance across five seeds, confirming the corrected pipeline behaves as intended.

We regard the most probable explanation as sampling frequency: at a median 18.2-second inter-update interval, the short-horizon dynamics that higher-frequency work exploits occur between our observations. If so, the binding constraint for retail participation is data access rather than modelling capability. That is a less encouraging conclusion than the one we set out to reach, and a more actionable one.

The framework of Section 3 remains, we believe, a correct formalisation of the problem. What we have shown is that its first viability condition fails in the specific setting where it would matter most to the participants it was designed to serve. Establishing whether it holds at finer data resolution is the natural next step, and requires only a feed we did not have.

Data and Code Availability

The complete collection pipeline, feature engineering code, models, ablation and baseline scripts, and result files are available at <https://github.com/azure-cyber/latency-compensating-mm>. All results in this paper are reproducible from that repository. The diagnostic and baseline scripts (`diagnose.py`, `baseline_check.py`, `ablation.py`, `per_asset.py`) are included so the null result can be checked directly.

Disclosure of AI Assistance

The author used Claude (Anthropic) extensively throughout this project: for implementation of the data collection pipeline, feature engineering, models, and evaluation code; for diagnostic analysis including identification of the conditioning defect described in Section 6; and for drafting and revision of this manuscript. The author designed the study, directed the analysis, made all substantive research decisions, verified the results, and is solely responsible for the content and any errors.

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